

Step 5: Requirements and Agile Setup (Sprint 0)

5.1 Agile Delivery Approach:

This project is delivered using Agile Scrum to enable incremental value delivery and early stakeholder feedback. Work is structured into a short Sprint 0 for discovery and setup, followed by sprints 1-4 focusing on forecasting, KPI measurement, dashboard delivery, and actionable recommendations. The approach prioritises an MVP that delivers basic forecasting accuracy tracking and fulfilment KPI visibility before adding enhancements.

5.2 Scrum Roles

Product Owner: Operations Manager (Prioritises outcomes: OTIF, stock-outs)

Scrum Master : Project lead (removes blockers, ensures cadence)

Delivery Team: Business Analyst, BI analyst, data analyst

5.3 MVP Definition:

The MVP is considered successful if it delivers:

1. A clean, documented dataset with agreed KPI definitions.
2. A forecast model with measured accuracy (MAPE)
3. A basic replenishment recommendation logic
4. A Power BI dashboard showing key fulfilment trends, On-Time Delivery, monthly order volumes, revenue, and decision-support visuals for operational monitoring.

order volumes, revenue, and decision support visuals for operational monitoring.

5. A weekly review cadence and an exception log

5.4 Epics:

Epic ID	Epic Name	Description	Business Value
E1	Data Foundation & Governance	Define data structure, cleaning rules, KPI definitions	Trustworthy decisions
E2	Forecasting & Accuracy Tracking	Build baseline + improved forecasts, track MAPE/MAD	Better planning
E3	Replenishment Decision Support	Translate forecast into reorder recommendations	Reduced stock-outs/overstock
E4	KPI Monitoring Dashboard	Design and deliver a Power BI dashboard for fulfilment and forecasting KPIs	Visibility & control
E5	Exception Management & Continuous Improvement	Define exception log, review cadence, improvement loop	Faster response

5.5 User Stories and Acceptance Criteria:

OTIF	As a	I want	So that	Acceptance Criteria(Given/when/Then)
US1	Supply Chain Planner	A standard dataset Template	Data is consistent	Given raw data when cleaned. Then missing

			across months	values are flagged and a clean table is produced
US2	BI Analyst	KPI Definitions documented	Dashboards are consistent	Given KPI list, when calculated, then formulas and definitions are recorded in a data dictionary
US3	Operations Manager	OTIF and Fill Rate tracking	I can monitor service	Given order data, when KPIs computed, then OTIF/Fill rate are available monthly by category
US4	Supply Chain Planner	Baseline forecast(native/moving average)	I have a benchmark	Given historical demand, when baseline methods run, then forecasts are produced for next period
US5	Supply Chain Planner	Exponential Smoothing Forecast	Accuracy improves	Given demand series, when alpha changes, then forecast updates and MAPE recalculates
US6	Operations Manager	Forecast Accuracy view	I can trust Planning	Given forecasts and actuals, when compared, then MAPE is shown over time
US7	Supply Chain Planner	A Simple replenishment Recommendation	Ordering is less reactive	Given forecast and lead time, when reorder point logic is applied, then suggested order quantity is generated
US8	Operations Manager	A Power BI Dashboard	Decisions are Faster	Given clean data, when loaded to Power BI, then KPI cards and trend visuals render correctly with filters.

US9	Operations Manager	An exception log	Issues are tracked	Given KPI threshold, when breached, then exception is recorded with date, reason, action
US10	Product Owner	Sprint demos	I can confirm values	Given sprint increment, when review happens, then working outputs are demonstrated and feedback is captured

5.6 Sprint Plan:

Sprint	Goal	Key Deliverables	Demo Output
--------	------	------------------	-------------

Sprint 0	Align on Scope & Setup	Charter, KPIs, backlog, data template	Ready to build pack
Sprint1	Baseline Forecasting	Clean dataset + baseline forecast + MAPE	Excel model v1
Sprint2	Improve forecasting + replenishment	Smoothing forecast + recommendation logic	Excel model v2

Sprint 3	Dashboard Delivery	Power BI dashboard + KPI trends	Dashboard demo
Sprint 4	Decision Support	Insights recommendations, implementation plan	Final Advisory Pack

5.7 Definition of Ready (DoR):

A story is “Ready” when: data is available, acceptance criteria written, dependencies known, estimated, and fits within sprint capacity.

5.8 Definition of Done (DoD):

A story is “Done” when: output is created, validated, documented, reviewed, and included in the sprint demo with evidence (screenshots/tables).

5.9 Risks & Dependencies:

- Dependency: clean data availability (Sprint 0)
- Risk: KPI mismatch between Excel and Power BI
- Mitigation: single KPI definition page + reconciliation checks

5.10 Requirements Traceability

Epic	User Stories	Output
E1	US1-US2	Clean dataset + KPI definitions
E2	US4-US6	Forecast model + accuracy
E3	US7	Replenishment logic
E4	US8	Power BI dashboard
E5	US9-US10	Exception log + sprint demos